

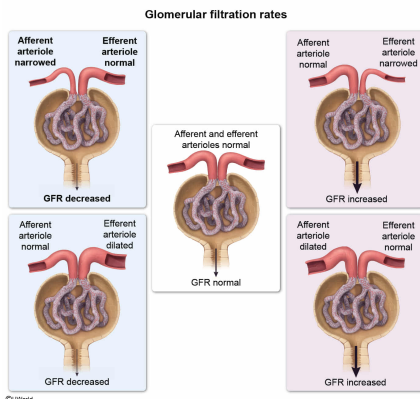
Stenosis is the abnormal narrowing of a tubular structure in the body, such as a blood vessel. Stenosis of the afferent arterioles in the kidney would be expected to:

- ✓ ☒ A. decrease the glomerular filtration rate. [79%]
☐ B. increase blood flow to the glomeruli. [12%]
☐ C. decrease resistance in the afferent arterioles. [3%]
☐ D. increase the output of urine. [5%]

Correct

78% answered correctly

Explanation:



Blood flow in the circulation resembles **current flow** in electrical circuits and is influenced by the resistance (ie, opposition to flow) of the vessels. Within *single vessels*, narrowing (eg, vasoconstriction, stenosis) leads to increased resistance and, in the absence of any other changes, reduced pressure according to **Bernoulli's equation**. Within the *circulatory system*, blood flows through *different vessels* in parallel (eg, arterioles within an organ) according to the **relative vessel resistances**. As with current flow through parallel resistors, low-resistance routes receive more blood flow than parallel high-resistance routes.

In the kidney's **nephrons**, circulating **blood enters** the **glomerulus** through the **afferent arteriole** and exits through the efferent arteriole. The degree of **vascular constriction** of these two arterioles **controls** the rate of **blood flow into and out of the glomerulus**, thereby affecting the volume of fluid filtered through the kidney per unit time, known as the **glomerular filtration rate (GFR)**.

Glomerular hydrostatic (blood) pressure contributes to blood **filtration**, forcing excess fluid and waste products (filtrate) across the porous endothelium and into Bowman's space. In general, proteins and cells are too large to pass through the pores, and the osmotic influence of proteins on fluid movement into and out of blood vessels is referred to as oncotic pressure.

According to the question, stenosis is an abnormal narrowing of a blood vessel, a condition that mimics the physiological effect of vasoconstriction. As a result, **afferent arteriole stenosis** would limit the amount of blood flow through the glomerulus, causing *decreased* glomerular hydrostatic pressure and, consequently, **decreased GFR**.

(Choices B and C) Stenosis (narrowing) of the afferent arteriole would likely *increase* (*not* decrease) resistance to blood flow through the vessel (ie, due to reduced vessel diameter). This would *decrease* (*not* increase) blood flow into the afferent arteriole and glomerulus.

(Choice D) Urine production is influenced by the GFR, the rate of reabsorption (movement of substances from the filtrate back into the bloodstream), and the rate of secretion (movement of substances from the bloodstream to the filtrate). When GFR decreases, as it does with afferent arteriole stenosis, less filtrate is produced and urine output *decreases*.

Educational objective:

The volume of fluid filtered through the kidney per unit time is known as the glomerular filtration rate (GFR), which can be modulated by blood flow through the glomerulus. Higher hydrostatic (blood) pressure in the glomerulus increases GFR, and lower hydrostatic pressure decreases GFR.